AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.062 / 0.0155032 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-19 20:00 UTC
Location	29.7644, -95.0781
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 680	37.3 to 100; mean 73.9437	50 at 2026-07-19 19:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 680	23.1111 to 37; mean 29.6197	35 at 2026-07-19 19:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 844	0 to 320; mean 164.171	100 at 2026-07-19 19:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 889	0 to 8.74555; mean 2.519	2.57222 at 2026-07-19 19:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 96	5919.26 to 5942.92; mean 5928.68	5926.1 at 2026-07-19 18:00Z; 2 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 96	33.3746 to 1860.78; mean 682.587	1296.31 at 2026-07-19 18:00Z; 2 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 96	29.7056 to 47.063; mean 36.7124	36.0384 at 2026-07-19 18:00Z; 2 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 96	1006.67 to 1018.53; mean 1012.63	1014.5 at 2026-07-19 18:00Z; 2 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 96	18.0655 to 23.3589; mean 21.5589	21.9215 at 2026-07-19 18:00Z; 2 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 96	-2.78129 to 2.90839; mean 0.7987	1.17125 at 2026-07-19 18:00Z; 2 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 96	-0.0770996 to 4.7729; mean 2.0657	1.0878 at 2026-07-19 18:00Z; 2 h before event
openaq	Ozone (ozone)	ppm	16 / 1059	-0.004 to 0.094; mean 0.0184	0.062 at 2026-07-19 20:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	2 / 144	29 to 66; mean 42.125	53 at 2026-07-19 20:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	73 / 3162	0.3 to 45.6; mean 7.8606	18.3 at 2026-07-19 20:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	4 / 285	0 to 100; mean 53.7544	75 at 2026-07-19 20:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	4 / 285	44 to 96; mean 76.2421	58 at 2026-07-19 20:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	4 / 285	0 to 0; mean 0	0 at 2026-07-19 20:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	4 / 285	1011 to 1020; mean 1015	1014 at 2026-07-19 20:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	4 / 285	24.26 to 37.22; mean 30.2321	35.32 at 2026-07-19 20:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	4 / 285	0 to 318; mean 197.13	107 at 2026-07-19 20:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	4 / 285	0.45 to 5.9; mean 2.1239	1.79 at 2026-07-19 20:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	64 / 4459	0 to 83.5; mean 9.2588	8.8 at 2026-07-19 20:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	4 / 4	0.0254212 to 0.0262849; mean 0.0259	0.0254212 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	7 / 7	0.000157838 to 0.000213688; mean 0.0002	0.000176599 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	1.64557e-05 to 5.83485e-05; mean 0	3.22877e-05 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	7 / 7	-3.60822e-05 to 0.000116366; mean 0	2.99088e-06 at 2026-07-19 20:09Z; 10 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 10 min after event
tceq	Carbon monoxide (co)	ppm	3 / 185	0 to 0.8; mean 0.2135	0.1 at 2026-07-19 20:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 690	-2.5 to 20.7; mean 3.9258	9.6 at 2026-07-19 20:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 179	-0.6 to 6.9; mean 0.0894	0.1 at 2026-07-19 20:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-18 06Z 90.58, 12Z 71, 18Z 58.04, 07-19 00Z 80.16, 06Z 92.78, 12Z 74.29, 18Z 53.98, 07-20 00Z 79.43, 06Z 93.45, 12Z 74.98, 18Z 48.76, 07-21 00Z 75.65, 06Z 88.41
asos	Air temperature (temperature)	07-18 06Z 26.6, 12Z 30.71, 18Z 33.1, 07-19 00Z 28.32, 06Z 25.57, 12Z 29.51, 18Z 34, 07-20 00Z 28.82, 06Z 25.33, 12Z 29.14, 18Z 34.4, 07-21 00Z 29.33, 06Z 25.95
asos	Wind direction (wind_direction)	07-18 06Z 127.4, 12Z 181.1, 18Z 184.7, 07-19 00Z 157.8, 06Z 84.44, 12Z 190.3, 18Z 139.6, 07-20 00Z 162.1, 06Z 145.7, 12Z 226, 18Z 159.2, 07-21 00Z 178.9, 06Z 224.6
asos	Wind speed (wind_speed)	07-18 06Z 1.967, 12Z 2.868, 18Z 3.985, 07-19 00Z 3.073, 06Z 0.9612, 12Z 2.417, 18Z 2.209, 07-20 00Z 3.021, 06Z 1.75, 12Z 2.545, 18Z 2.209, 07-21 00Z 2.984, 06Z 2.594
noaa_gfs	500-hPa geopotential height (gh_500)	07-18 12Z 5938, 18Z 5943, 07-19 00Z 5930, 06Z 5936, 12Z 5923, 18Z 5926, 07-20 00Z 5922, 06Z 5930, 12Z 5920, 18Z 5928, 07-21 00Z 5923, 06Z 5926
noaa_gfs	Planetary boundary layer height (pbl_height)	07-18 12Z 279.6, 18Z 1616, 07-19 00Z 874.3, 06Z 339.5, 12Z 180.6, 18Z 1341, 07-20 00Z 798.1, 06Z 247.3, 12Z 149.4, 18Z 1284, 07-21 00Z 856.6, 06Z 224.3
noaa_gfs	Precipitable water (precipitable_water)	07-18 12Z 41.38, 18Z 45.04, 07-19 00Z 38.96, 06Z 36.45, 12Z 34.75, 18Z 36.26, 07-20 00Z 39.58, 06Z 34.69, 12Z 30.31, 18Z 32.87, 07-21 00Z 36.08, 06Z 34.18
noaa_gfs	Surface pressure (surface_pressure)	07-18 12Z 1017, 18Z 1016, 07-19 00Z 1013, 06Z 1014, 12Z 1014, 18Z 1013, 07-20 00Z 1011, 06Z 1012, 12Z 1012, 18Z 1012, 07-21 00Z 1009, 06Z 1009
noaa_gfs	850-hPa temperature (t_850)	07-18 12Z 20.79, 18Z 18.49, 07-19 00Z 20.69, 06Z 22.5, 12Z 21.94, 18Z 21.4, 07-20 00Z 20.94, 06Z 22.66, 12Z 22.62, 18Z 22.26, 07-21 00Z 21.38, 06Z 23.02
noaa_gfs	10-m eastward wind (u_10m)	07-18 12Z 0.768, 18Z 0.2137, 07-19 00Z -0.389, 06Z 1.243, 12Z 0.7484, 18Z 1.356, 07-20 00Z -0.8788, 06Z 1.619, 12Z 1.532, 18Z 1.818, 07-21 00Z -0.3229, 06Z 1.877
noaa_gfs	10-m northward wind (v_10m)	07-18 12Z 1.805, 18Z 2.916, 07-19 00Z 4.105, 06Z 2.243, 12Z 1.512, 18Z 1.434, 07-20 00Z 2.877, 06Z 2.366, 12Z 0.4644, 18Z 0.5732, 07-21 00Z 2.195, 06Z 2.3
openaq	Ozone (ozone)	07-18 06Z 0.01095, 12Z 0.01519, 18Z 0.02561, 07-19 00Z 0.01543, 06Z 0.008458, 12Z 0.01438, 18Z 0.03453, 07-20 00Z 0.01715, 06Z 0.006741, 12Z 0.01062, 18Z 0.04376, 07-21 00Z 0.01854, 06Z 0.007138
openaq	PM10 (pm10)	07-18 06Z 35.12, 12Z 39.5, 18Z 37.18, 07-19 00Z 39.33, 06Z 35.58, 12Z 35.75, 18Z 46.83, 07-20 00Z 45.25, 06Z 44.75, 12Z 49.67, 18Z 50.75, 07-21 00Z 43.55, 06Z 41.33
openaq	PM2.5 (pm25)	07-18 06Z 9.241, 12Z 8.502, 18Z 5.846, 07-19 00Z 7.481, 06Z 8.598, 12Z 6.654, 18Z 8.287, 07-20 00Z 7.733, 06Z 8.757, 12Z 8.054, 18Z 6.761, 07-21 00Z 6.957, 06Z 7.602
openweather	Cloud cover (cloud_cover)	07-18 06Z 99.44, 12Z 88.36, 18Z 98.83, 07-19 00Z 97.96, 06Z 92.96, 12Z 92.95, 18Z 81.62, 07-20 00Z 12.04, 06Z 5.64, 12Z 2.958, 18Z 3.208, 07-21 00Z 3.739, 06Z 1
openweather	Relative humidity (humidity)	07-18 06Z 90.31, 12Z 79.48, 18Z 63.67, 07-19 00Z 77.74, 06Z 90.68, 12Z 81.62, 18Z 60.38, 07-20 00Z 75.61, 06Z 90.72, 12Z 80.17, 18Z 53.25, 07-21 00Z 72.3, 06Z 84.88
openweather	Precipitation rate (precipitation)	07-18 06Z 0, 12Z 0, 18Z 0, 07-19 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-20 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-21 00Z 0, 06Z 0
openweather	Surface pressure (pressure)	07-18 06Z 1018, 12Z 1019, 18Z 1016, 07-19 00Z 1016, 06Z 1016, 12Z 1017, 18Z 1014, 07-20 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1012, 07-21 00Z 1012, 06Z 1012
openweather	Air temperature (temperature)	07-18 06Z 26.82, 12Z 29.86, 18Z 34.21, 07-19 00Z 29.55, 06Z 26.26, 12Z 28.92, 18Z 35.34, 07-20 00Z 30.51, 06Z 25.94, 12Z 28.87, 18Z 35.82, 07-21 00Z 30.83, 06Z 26.99
openweather	Wind direction (wind_direction)	07-18 06Z 177.6, 12Z 196.7, 18Z 180, 07-19 00Z 182.6, 06Z 192.6, 12Z 212.5, 18Z 186.3, 07-20 00Z 181.8, 06Z 186.9, 12Z 256, 18Z 221.3, 07-21 00Z 184.5, 06Z 200
openweather	Wind speed (wind_speed)	07-18 06Z 1.737, 12Z 2.018, 18Z 2.075, 07-19 00Z 2.62, 06Z 1.796, 12Z 1.75, 18Z 2.181, 07-20 00Z 2.799, 06Z 1.84, 12Z 2.049, 18Z 1.876, 07-21 00Z 2.613, 06Z 2.296

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-18 06Z 11.34, 12Z 9.617, 18Z 7.394, 07-19 00Z 9.028, 06Z 10.5, 12Z 8.548, 18Z 8.768, 07-20 00Z 9.176, 06Z 10.87, 12Z 9.812, 18Z 7.692, 07-21 00Z 8.914, 06Z 9.637
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	CO column (s5p_co_column)	07-19 18Z 0.02551, 07-20 18Z 0.02628
sentinel5p	CO granules available (s5p_co_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-18 18Z 0.0001745, 07-19 18Z 0.0001766, 07-20 18Z 0.000213
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-18 18Z 1.872e-05, 07-19 18Z 3.229e-05, 07-20 18Z 5.835e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-18 18Z 3.278e-05, 07-19 18Z 2.991e-06, 07-20 18Z 1.73e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
tceq	Carbon monoxide (co)	07-18 06Z 0.225, 12Z 0.2556, 18Z 0.2556, 07-19 00Z 0.25, 06Z 0.1944, 12Z 0.2111, 18Z 0.2, 07-20 00Z 0.2667, 06Z 0.2056, 12Z 0.2125, 18Z 0.1722, 07-21 00Z 0.2222, 06Z 0.08889
tceq	Nitrogen dioxide (no2)	07-18 06Z 3.966, 12Z 3.301, 18Z 2.128, 07-19 00Z 1.933, 06Z 3.03, 12Z 2.541, 18Z 3.297, 07-20 00Z 3.545, 06Z 4.577, 12Z 6.21, 18Z 6.136, 07-21 00Z 6.723, 06Z 4.179
tceq	Sulfur dioxide (so2)	07-18 06Z -0.3583, 12Z -0.3222, 18Z -0.2889, 07-19 00Z -0.1, 06Z -0.4533, 12Z 0.1056, 18Z 0.4389, 07-20 00Z -0.2875, 06Z -0.3471, 12Z -0.175, 18Z 1.906, 07-21 00Z 1.213, 06Z -0.2556

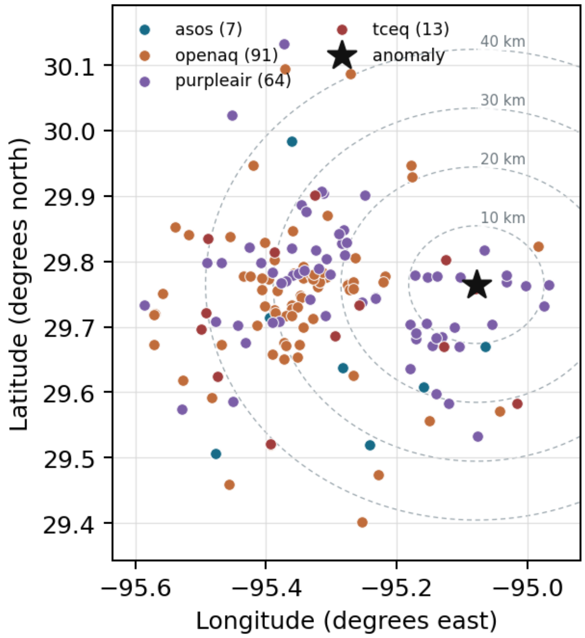
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

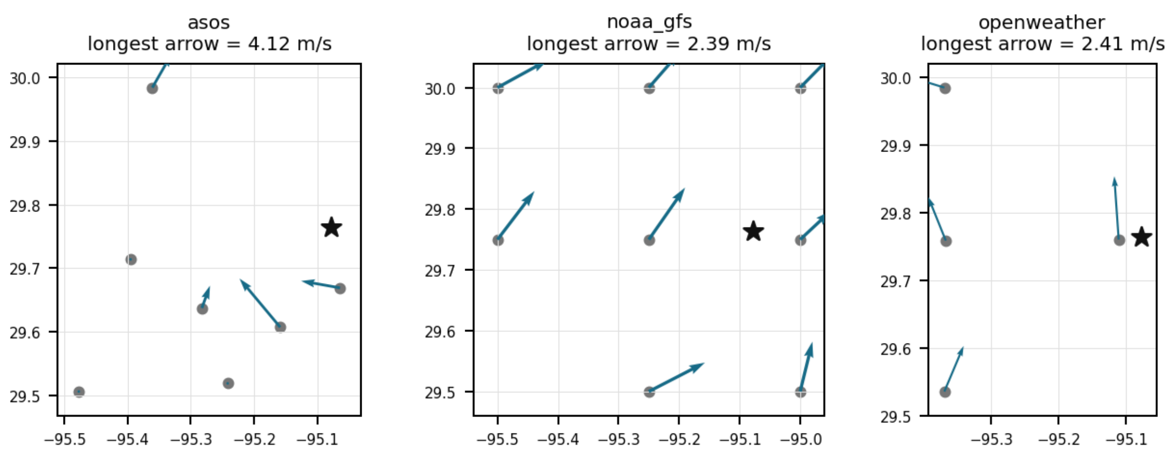
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (10.672 km) 75.56, EFD (19.125 km) 74.67, HOU (24.259 km) 74.61, MCJ (31.106 km) 72.29, LVJ (31.547 km) 75.46, IAH (36.618 km) 72.99, AXH (48.068 km) 73.14
asos	Air temperature (temperature)	T41 (10.672 km) 29.67, EFD (19.125 km) 29.86, HOU (24.259 km) 29.65, MCJ (31.106 km) 29.83, LVJ (31.547 km) 29.25, IAH (36.618 km) 29.99, AXH (48.068 km) 29.21
asos	Wind direction (wind_direction)	T41 (10.672 km) 188.4, EFD (19.125 km) 169.1, HOU (24.259 km) 196, MCJ (31.106 km) 174.1, LVJ (31.547 km) 140.7, IAH (36.618 km) 170.6, AXH (48.068 km) 113.4
asos	Wind speed (wind_speed)	T41 (10.672 km) 3.324, EFD (19.125 km) 3.113, HOU (24.259 km) 3.205, MCJ (31.106 km) 3.173, LVJ (31.547 km) 1.822, IAH (36.618 km) 2.244, AXH (48.068 km) 1.037
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.00 (7.707 km) 5929, gfs:29.75,-95.25 (16.671 km) 5929, gfs:30.00,-95.00 (27.258 km) 5928, gfs:29.50,-95.00 (30.354 km) 5929, gfs:30.00,-95.25 (31.0 km) 5929, gfs:29.50,-95.25 (33.77 km) 5929, gfs:29.75,-95.50 (40.758 km) 5929, gfs:30.00,-95.50 (48.382 km) 5929
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.00 (7.707 km) 623.9, gfs:29.75,-95.25 (16.671 km) 770.6, gfs:30.00,-95.00 (27.258 km) 446.7, gfs:29.50,-95.00 (30.354 km) 540.1, gfs:30.00,-95.25 (31.0 km) 846.2, gfs:29.50,-95.25 (33.77 km) 470.7, gfs:29.75,-95.50 (40.758 km) 885.4, gfs:30.00,-95.50 (48.382 km) 877.2

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.00 (7.707 km) 37.5, gfs:29.75,-95.25 (16.671 km) 36.71, gfs:30.00,-95.00 (27.258 km) 37.41, gfs:29.50,-95.00 (30.354 km) 37.29, gfs:30.00,-95.25 (31.0 km) 36.56, gfs:29.50,-95.25 (33.77 km) 36.75, gfs:29.75,-95.50 (40.758 km) 35.91, gfs:30.00,-95.50 (48.382 km) 35.57
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.00 (7.707 km) 1014, gfs:29.75,-95.25 (16.671 km) 1013, gfs:30.00,-95.00 (27.258 km) 1012, gfs:29.50,-95.00 (30.354 km) 1015, gfs:30.00,-95.25 (31.0 km) 1012, gfs:29.50,-95.25 (33.77 km) 1013, gfs:29.75,-95.50 (40.758 km) 1012, gfs:30.00,-95.50 (48.382 km) 1010
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.00 (7.707 km) 21.68, gfs:29.75,-95.25 (16.671 km) 21.59, gfs:30.00,-95.00 (27.258 km) 21.53, gfs:29.50,-95.00 (30.354 km) 21.84, gfs:30.00,-95.25 (31.0 km) 21.31, gfs:29.50,-95.25 (33.77 km) 21.71, gfs:29.75,-95.50 (40.758 km) 21.37, gfs:30.00,-95.50 (48.382 km) 21.44
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.00 (7.707 km) 0.3742, gfs:29.75,-95.25 (16.671 km) 0.8576, gfs:30.00,-95.00 (27.258 km) 0.3609, gfs:29.50,-95.00 (30.354 km) 0.8509, gfs:30.00,-95.25 (31.0 km) 0.6184, gfs:29.50,-95.25 (33.77 km) 0.6842, gfs:29.75,-95.50 (40.758 km) 1.107, gfs:30.00,-95.50 (48.382 km) 1.537
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.00 (7.707 km) 1.981, gfs:29.75,-95.25 (16.671 km) 2.176, gfs:30.00,-95.00 (27.258 km) 2.046, gfs:29.50,-95.00 (30.354 km) 2.674, gfs:30.00,-95.25 (31.0 km) 1.659, gfs:29.50,-95.25 (33.77 km) 2.362, gfs:29.75,-95.50 (40.758 km) 2.046, gfs:30.00,-95.50 (48.382 km) 1.583
openaq	Ozone (ozone)	2800 (0.0 km) 0.0214, 3214 (6.242 km) 0.02314, 339 (11.214 km) 0.02646, 332 (11.568 km) 0.02039, 2039 (13.761 km) 0.02115, 3378 (17.648 km) 0.01379, 5077791 (21.045 km) 0.01486, 2046 (21.111 km) 0.01903, +8 more
openaq	PM10 (pm10)	271 (11.568 km) 47.54, 15852419 (23.148 km) 36.86
openaq	PM2.5 (pm25)	268 (11.568 km) 16.99, 13787195 (13.589 km) 10.33, 15539682 (17.663 km) 9.821, 8967123 (17.705 km) 11.39, 8967479 (17.705 km) 10.84, 9489522 (18.242 km) 8.637, 9201803 (18.26 km) 9.998, 14157976 (18.483 km) 9.464, +65 more
openweather	Cloud cover (cloud_cover)	grid:east (3.216 km) 53.69, grid:center (27.961 km) 52.52, grid:north (37.319 km) 54.94, grid:south (37.957 km) 53.86
openweather	Relative humidity (humidity)	grid:east (3.216 km) 80.38, grid:center (27.961 km) 74.46, grid:north (37.319 km) 74.59, grid:south (37.957 km) 75.54
openweather	Precipitation rate (precipitation)	grid:east (3.216 km) 0, grid:center (27.961 km) 0, grid:north (37.319 km) 0, grid:south (37.957 km) 0
openweather	Surface pressure (pressure)	grid:east (3.216 km) 1015, grid:center (27.961 km) 1015, grid:north (37.319 km) 1015, grid:south (37.957 km) 1015
openweather	Air temperature (temperature)	grid:east (3.216 km) 29.87, grid:center (27.961 km) 30.74, grid:north (37.319 km) 30.34, grid:south (37.957 km) 29.98
openweather	Wind direction (wind_direction)	grid:east (3.216 km) 200.8, grid:center (27.961 km) 212.6, grid:north (37.319 km) 174.3, grid:south (37.957 km) 200.8
openweather	Wind speed (wind_speed)	grid:east (3.216 km) 2.642, grid:center (27.961 km) 1.77, grid:north (37.319 km) 1.783, grid:south (37.957 km) 2.298
purpleair	PM2.5 (pm25)	148709 (2.751 km) 10.92, 213701 (4.519 km) 9.659, 268171 (4.647 km) 8.861, 186271 (5.953 km) 10.6, 186277 (5.959 km) 11.13, 222815 (7.168 km) 9.082, 166427 (7.312 km) 12.82, 166441 (7.414 km) 8.636, +56 more
tceq	Carbon monoxide (co)	482011039 (11.598 km) 0.141, 482011035 (17.662 km) 0.1317, 482011052 (30.401 km) 0.3594
tceq	Nitrogen dioxide (no2)	482011015 (0.618 km) 5.302, 482010026 (6.25 km) 4.844, 482011039 (11.598 km) 2.742, 482011035 (17.662 km) 5.613, 482011050 (21.051 km) 0.8661, 482010416 (22.65 km) 4.593, 482010024 (28.341 km) 2.079, 482011052 (30.401 km) 7.826, +4 more
tceq	Sulfur dioxide (so2)	482011039 (11.598 km) 0.2525, 482011035 (17.662 km) -0.2576, 482010051 (41.337 km) 0.2672

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

Elevated tropospheric NO_2 column densities and local ground-level NO_2 concentrations provided key precursors necessary for rapid photochemical ozone production.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

Local nitrogen dioxide concentrations reached 9.6 ppb near the anomaly site, supplying necessary precursors for photochemical ozone formation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to the accumulation of pollutants in the air.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

Hot afternoon conditions supported rapid ozone photochemistry in eastern Houston because air temperature near the anomaly was about 35.0 to 35.3 degC at 19:55 to 20:02 UTC, humidity dropped to about 50 to 58 percent, and ozone across the openaq network increased to a 18Z six-hour mean of 0.03453 ppm while the anomalous site reached 0.062 ppm at 20:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

Elevated tropospheric NO2 column densities observed by Sentinel-5P alongside ground-level NO2 concentrations detected by TCEQ monitors provided essential nitrogen dioxide precursors for photochemical ozone synthesis.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

Transport of NOx-rich urban or industrial precursor air into the eastern Houston receptor was likely a contributor on 2026-07-19 20:00 UTC because local wind direction was easterly to east-southeasterly near the anomaly time and nearby NO2 was elevated relative to the broader period while CO and SO2 remained low, favoring ozone formation from fresh precursor plumes rather than combustion smoke or a sulfur-dominated release.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

Local photochemical ozone production was likely enhanced near eastern Houston on 2026-07-19 20:00 UTC because surface temperatures were very high at about 35 C, humidity had dropped to about 50 to 58 percent, and no precipitation was occurring during the afternoon ozone rise.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

A severe thunderstorm or tornado event has stirred up dust and pollutants from the ground, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

OpenAQ reported a severe ozone anomaly reaching 0.062 ppm at location (29.764, -95.078) on 2026-07-19T20:00:00+00:00, compared to an expected value of approximately 0.0155 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

Surface temperatures peaked at 35.0 degC alongside low relative humidity around 50 percent during the afternoon of July 19, 2026, driving strong photochemical activity.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

Meteorological conditions near eastern Houston at the anomaly time favored ozone buildup because temperatures were about 35 C, humidity had dropped to roughly 50 to 60 percent, winds were light at about 2 m/s, and the daytime boundary layer was moderately deep rather than strongly ventilated.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

Meteorological dispersion was only moderate near eastern Houston on 2026-07-19 20:00 UTC because surface winds were light at about 1.8 to 2.6 m/s while boundary-layer height was elevated near 1300 m, a combination consistent with retention and daytime mixing of ozone precursors rather than strong flushing.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

The openaq ozone anomaly in eastern Houston at 2026-07-19 20:00 UTC is most consistent with a daytime photochemical ozone production episode under hot afternoon conditions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

The anomaly occurred on July 19th, 2026, around 20:00 UTC, and was located approximately 10.87 km from the nearest sensor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

The eastern Houston ozone spike is more likely linked to transport and aging of urban-industrial precursor emissions than to wildfire smoke or a sulfur-rich combustion event because NO2 was present near the site while CO, SO2, satellite CO, and particulate matter did not show signatures of a major smoke or sulfur plume.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

The openaq ozone network within 50 km had a 6-hour mean of 0.03453 ppm for 2026-07-19 18Z, which is lower than the 0.062 ppm value measured at the anomaly location at 20:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

The PM2.5 levels are elevated at 8.8 ug/m3, which is higher than the mean of 9.2588 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

An ozone concentration anomaly reaching 0.062 ppm was detected on July 19, 2026 at 20:00 UTC in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

ASOS and OpenWeather surface weather observations near the anomaly at 2026-07-19T20:00:00+00:00 showed high surface temperatures between 35.0 degC and 35.32 degC along with decreased relative humidity between 50.0 percent and 58.0 percent.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The openaq anomaly detection labeled the 2026-07-19T20:00:00+00:00 ozone event at 29.764, -95.078 as severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

The openaq ozone anomaly at coordinates 29.764, -95.078 occurred at 2026-07-19T20:00:00+00:00 with an observed value of 0.062 ppm, an expected value of 0.015503184713375794 ppm, and a z-score of 6.698251307179393.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

Light surface winds averaging approximately 2.2 m/s constrained atmospheric transport and dispersion, facilitating the local accumulation of ozone photoproducts.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

TCEQ ground monitors near the anomaly site recorded nitrogen dioxide concentrations up to 9.6 ppb at 2026-07-19T20:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

Only modest dispersion likely helped ozone accumulate near the receptor because surface wind speed near the anomaly was weak at about 1.8 to 2.6 m/s around 20:00 UTC while boundary-layer height was elevated near 1296 m, a combination consistent with enough mixing for photochemical production but not strong ventilation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

High ambient surface temperatures reaching 35.0 degC during peak afternoon hours accelerated the photochemical reaction rate of ozone formation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

A transported urban or industrial precursor plume is more plausible than a smoke-rich or sulfur-rich plume because near-anomaly NO_2 was elevated to 9.6 ppb at 20:00 UTC and the sentinel5p NO_2 column on 2026-07-19 exceeded the previous day, while CO remained low near 0.1 ppm, PM2.5 stayed modest near 8.8 to 18.3 ug/m3, and SO_2 remained very low near 0.1 ppb locally and near zero in the satellite column.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO and NO2 emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Light surface wind speeds around 2.2 to 2.6 m/s recorded by ASOS stations limited atmospheric transport and ventilation, facilitating the local accumulation of ozone near the monitoring site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

The air quality anomaly in Houston, Texas on July 19th is characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

High afternoon surface temperatures reaching 35°C to 37°C and low relative humidity near 50% created optimal thermodynamic conditions for rapid photochemical ozone formation during peak solar hours.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

The air quality anomaly in Houston, Texas on July 19th is also characterized by elevated levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

There is no evidence to suggest that the air quality anomaly in Houston, Texas on July 19th was caused by any specific weather or atmospheric conditions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.